Superconducting Hartree-Fock mean-field theory

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I. TIME-DEPENDENT HARTREE-FOCK MEAN-FIELD THEORY

A. Introduction

Consider an equilibirum many-body system of electrons described by the Hamiltonian

$$H_{\rm eq} = H_0 + H_{\rm int} = \sum_{\alpha\beta} h_{\alpha\beta} c^\dagger_\alpha c_\beta + \sum_{\alpha\beta\gamma\delta} V^{\alpha\beta}_{\gamma\delta} c^\dagger_\alpha c^\dagger_\beta c_\gamma c_\delta, \tag{1}$$

with $c^\dagger(c)$ fermionic creation (annihilation) operators and the Greek indices representing generic degrees of freedom. Concerning the interaction term, we assume $V(\mathbf{r})$ to be generic electrostatic potential of symmetry $V(\mathbf{r}-\mathbf{r}') = V(\mathbf{r}'-\mathbf{r})$ such that the interaction matrix elements have the symmetries

$$V^{\alpha\beta}_{\gamma\delta} = V^{\beta\alpha}_{\delta\gamma} \quad \text{and} \quad \left(V^{\alpha\beta}_{\gamma\delta}\right)^* = V^{\gamma\delta}_{\alpha\beta}. \tag{2}$$

Symmetry properties of V
From the imposed symmetry $V(\mathbf{r}-\mathbf{r}') = V(\mathbf{r}'-\mathbf{r})$, we obtain

$$\begin{aligned} V^{\alpha\beta}_{\gamma\delta} &= \int d\mathbf{r}d\mathbf{r}' \phi^*_\alpha(\mathbf{r})\phi^*_\beta(\mathbf{r}')V(\mathbf{r}-\mathbf{r}')\phi_\gamma(\mathbf{r}')\phi_\delta(\mathbf{r}),\ \mathbf{r} \leftrightharpoons \mathbf{r}' \\ &= \int d\mathbf{r}d\mathbf{r}' \phi^*_\alpha(\mathbf{r}')\phi^*_\beta(\mathbf{r})V(\mathbf{r}-\mathbf{r}')\phi_\gamma(\mathbf{r})\phi_\delta(\mathbf{r}') \\ &= \int d\mathbf{r}d\mathbf{r}' \phi^*_\beta(\mathbf{r})\phi^*_\alpha(\mathbf{r}')V(\mathbf{r}-\mathbf{r}')\phi_\delta(\mathbf{r}')\phi_\gamma(\mathbf{r}) = V^{\beta\alpha}_{\delta\gamma} \ \square \end{aligned}$$

Similarly, we can show that

$$\begin{aligned} \left(V^{\alpha\beta}_{\gamma\delta}\right)^* &= \int d\mathbf{r}d\mathbf{r}' \phi_\alpha(\mathbf{r})\phi_\beta(\mathbf{r}')V(\mathbf{r}-\mathbf{r}')\phi^*_\gamma(\mathbf{r}')\phi^*_\delta(\mathbf{r}) \\ &= \int d\mathbf{r}d\mathbf{r}' \phi^*_\delta(\mathbf{r})\phi^*_\gamma(\mathbf{r}')V(\mathbf{r}-\mathbf{r}')\phi_\delta(\mathbf{r}')\phi_\alpha(\mathbf{r}) = V^{\delta\gamma}_{\beta\alpha} = V^{\gamma\delta}_{\alpha\beta} \ \square \end{aligned}$$

Consider now that we perturb our system with an externally applied time-dependent force $F(t)$, such that system is instead described by the Hamiltonian

$$H(t) = H_{\rm eq} + H_{\rm per}(t) = \sum_{\alpha\beta} \left(h_{\alpha\beta} + B_{\alpha\beta}F(t)\right) c^\dagger_\alpha c_\beta + \frac{1}{2}\sum_{\alpha\beta\gamma\delta} V^{\alpha\beta}_{\gamma\delta} c^\dagger_\alpha c^\dagger_\beta c_\gamma c_\delta. \tag{3}$$

where $B_{\alpha\beta}$ are matrix elements of a one-particle operator that couples to the external force.

In order to study the dynamics of this system, we focus the system's time-dependent reduced density matrix (rDM) which is defined as

$$\boldsymbol{\rho}_{ba}(t) = \left\langle c^\dagger_a(t) c_b(t) \right\rangle. \tag{4}$$

Our work is thus to study the time-evolution of the rDM by solving its equation of motion (EoM).

B. Nambu representation

However, since we are looking for superconducting solutions, we wish to account also for the anomalous terms $\left\langle c^\dagger_\alpha c^\dagger_\beta \right\rangle$ and $\langle c_\gamma c_a \rangle$. For this, we introduce a Nambu representation as in Ref.(Nambu, 1959) where we consider the fermionic operators to be defined as spinors in spin space as

$$\boldsymbol{c}_\alpha(t) \equiv \begin{bmatrix} c_\alpha(t) \\ c^\dagger_\alpha(t) \end{bmatrix}, \tag{5}$$

The Nambu representation of the rDM can then be written as the tensor product of this spinors for each of the spin combinations,

$$\boldsymbol{\rho}_{ab} \equiv \left\langle \boldsymbol{c}_b^\dagger(t) \otimes \boldsymbol{c}_a(t) \right\rangle \equiv \begin{bmatrix} \left\langle c_b^\dagger c_a \right\rangle & \langle c_b c_a \rangle \\ \left\langle c_b^\dagger c_a^\dagger \right\rangle & \langle c_b c_a^\dagger \rangle \end{bmatrix} \equiv \begin{bmatrix} \rho_{ab}^{\mathrm{ee}} & \rho_{ab}^{\mathrm{eh}} \\ \rho_{ab}^{\mathrm{he}} & \rho_{ab}^{\mathrm{hh}} \end{bmatrix} \tag{6}$$

and omitted the time-dependency in both the rDM and fermionic operators for clarity. These terms are related by

$$\rho_{ab}^{\mathrm{hh}} = -\left(\rho_{ab}^{\mathrm{ee}}\right)^\dagger + \delta_{ab} \quad \text{and} \quad \rho_{ab}^{\mathrm{he}} = -\left(\rho_{ab}^{\mathrm{eh}}\right)^\dagger \tag{7}$$

For a BCS superconductor, i.e s-wave and spin-singlet, the pairing terms must have opposing spin however this is not the only possible type of superconductivity and thus, to keep the problem as general as possible we account for all the four terms. Read and write more about it. Our task now is then to solve the four (actually two, really) equation of motions for each of the terms in Eq.(6). In order to achieve a Nambu-symmetric equation of motion, we must manipulate the Hamiltonian in Eq.(**??**) and make a doubling of the degrees of freedom. We re-write it as

$$H(t) = \frac{1}{2}\sum_{\alpha\beta} \bar{h}_{\alpha\beta}\left(c_\alpha^\dagger c_\beta - c_\beta c_\alpha^\dagger\right) + \frac{1}{4}\sum_{\alpha\beta\gamma\delta} V_{\gamma\delta}^{\alpha\beta}\left(c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta + c_\delta c_\gamma c_\beta^\dagger c_\alpha^\dagger\right) + \frac{1}{2}\sum_\alpha \tilde{h}_{\alpha\alpha} \tag{8}$$

where we defined the quantity $\bar{h}_{\alpha\beta}(t) \equiv \tilde{h}_{\alpha\beta}(t) + \mu_{\alpha\beta}$ with the single-particle term $\tilde{h}$ and the chemical potential μ being

$$\tilde{h}_{\alpha\beta}(t) \equiv h_{\alpha\beta} + B_{\alpha\beta}F(t) \quad \text{and} \quad \mu_{\alpha\beta} = \frac{1}{2}\sum_\gamma \left(V_{\beta\gamma}^{\gamma\alpha} - V_{\gamma\beta}^{\gamma\alpha}\right) \tag{9}$$

Nambu-symmetrization of H

Concerning the single-particle terms we can manipulate the following expression as

$$c_\alpha^\dagger c_\beta = \frac{1}{2}c_\alpha^\dagger c_\beta + \frac{1}{2}c_\alpha^\dagger c_\beta = \frac{1}{2}c_\alpha^\dagger c_\beta + \frac{1}{2}\left(-c_\beta c_\alpha^\dagger + \delta_{\alpha\beta}\right) = \frac{1}{2}\left(c_\alpha^\dagger c_\beta - c_\beta c_\alpha^\dagger\right) + \frac{1}{2}\delta_{\alpha\beta} \tag{10}$$

Concerning the interacting terms, we can manipulate the following expression as

$$\begin{aligned} c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta &= c_\beta^\dagger c_\alpha^\dagger c_\delta c_\gamma \\ &= -c_\beta^\dagger c_\delta c_\alpha^\dagger c_\gamma + c_\beta^\dagger c_\gamma \delta_{\delta\alpha} \\ &= c_\delta c_\beta^\dagger c_\alpha^\dagger c_\gamma - c_\alpha^\dagger c_\gamma \delta_{\beta\delta} + c_\beta^\dagger c_\gamma \delta_{\delta\alpha} \\ &= -c_\delta c_\beta^\dagger c_\gamma c_\alpha^\dagger + c_\delta c_\beta^\dagger \delta_{\gamma\alpha} - c_\alpha^\dagger c_\gamma \delta_{\beta\delta} + c_\beta^\dagger c_\gamma \delta_{\delta\alpha} \\ &= c_\delta c_\gamma c_\beta^\dagger c_\alpha^\dagger - c_\delta c_\alpha^\dagger \delta_{\gamma\beta} + c_\delta c_\beta^\dagger \delta_{\gamma\alpha} - c_\alpha^\dagger c_\gamma \delta_{\beta\delta} + c_\beta^\dagger c_\gamma \delta_{\delta\alpha} \end{aligned} \tag{11}$$

allowing us to write

$$\begin{aligned} c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta =& \frac{1}{2}c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta + \frac{1}{2}c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta \\ =& \frac{1}{2}\left(c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta + c_\delta c_\gamma c_\beta^\dagger c_\alpha^\dagger\right) + \frac{1}{2}\left(c_\beta^\dagger c_\gamma \delta_{\delta\alpha} + c_\delta c_\beta^\dagger \delta_{\gamma\alpha} - c_\alpha^\dagger c_\gamma \delta_{\beta\delta} - c_\delta c_\alpha^\dagger \delta_{\gamma\beta}\right) \end{aligned} \tag{12}$$

Concerning ourselves with only the last four single particle terms, when put together with the interaction potential read as

$$
\begin{aligned}
&\frac{1}{4}\sum_{\alpha\beta\gamma\delta} V_{\gamma\delta}^{\alpha\beta}\left(c_\beta^\dagger c_\gamma \delta_{\delta\alpha} + c_\delta c_\beta^\dagger \delta_{\gamma\alpha} - c_\alpha^\dagger c_\gamma \delta_{\beta\delta} - c_\delta c_\alpha^\dagger \delta_{\gamma\beta}\right)\\
=&\frac{1}{4}\sum_{\alpha\beta\gamma}\left[V_{\gamma\alpha}^{\alpha\beta} c_\beta^\dagger c_\gamma + V_{\alpha\gamma}^{\alpha\beta} c_\gamma c_\beta^\dagger - V_{\gamma\beta}^{\alpha\beta} c_\alpha^\dagger c_\gamma - V_{\beta\gamma}^{\alpha\beta} c_\gamma c_\alpha^\dagger\right]\\
=&\frac{1}{4}\sum_{\alpha\beta\gamma}\left[V_{\gamma\alpha}^{\alpha\beta} c_\beta^\dagger c_\gamma + V_{\alpha\gamma}^{\alpha\beta}\left(-c_\beta^\dagger c_\gamma + \delta_{\beta\gamma}\right) - V_{\gamma\beta}^{\alpha\beta} c_\alpha^\dagger c_\gamma - V_{\beta\gamma}^{\alpha\beta}\left(-c_\alpha^\dagger c_\gamma + \delta_{\alpha\gamma}\right)\right]\\
=&\frac{1}{4}\sum_{\alpha\beta\gamma}\left[\left(V_{\gamma\alpha}^{\alpha\beta} - V_{\alpha\gamma}^{\alpha\beta}\right) c_\beta^\dagger c_\gamma + \left(V_{\beta\gamma}^{\alpha\beta} - V_{\gamma\beta}^{\alpha\beta}\right) c_\alpha^\dagger c_\gamma\right] + \frac{1}{4}\sum_{\alpha\beta}\left[V_{\alpha\beta}^{\alpha\beta} - V_{\beta\alpha}^{\alpha\beta}\right]\\
=&\frac{1}{2}\sum_{\alpha\beta\gamma}\left(V_{\beta\gamma}^{\gamma\alpha} - V_{\gamma\beta}^{\gamma\alpha}\right) c_\alpha^\dagger c_\beta + \frac{1}{4}\sum_{\alpha\beta}\left(V_{\alpha\beta}^{\alpha\beta} - V_{\beta\alpha}^{\alpha\beta}\right)\\
=&\frac{1}{4}\sum_{\alpha\beta\gamma}\left(V_{\beta\gamma}^{\gamma\alpha} - V_{\gamma\beta}^{\gamma\alpha}\right)\left[\left(c_\alpha^\dagger c_\beta - c_\beta c_\alpha^\dagger\right) + \delta_{\alpha\beta}\right] + \frac{1}{4}\sum_{\alpha\beta}\left(V_{\alpha\beta}^{\alpha\beta} - V_{\beta\alpha}^{\alpha\beta}\right)\\
=&\frac{1}{4}\sum_{\alpha\beta\gamma}\left(V_{\beta\gamma}^{\gamma\alpha} - V_{\gamma\beta}^{\gamma\alpha}\right)\left(c_\alpha^\dagger c_\beta - c_\beta c_\alpha^\dagger\right)
\end{aligned}
$$

The full expression for the interaction term is then

$$
H_{\text{int}} = \frac{1}{4}\sum_{\alpha\beta\gamma\delta} V_{\gamma\delta}^{\alpha\beta}\left(c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta + c_\delta c_\gamma c_\beta^\dagger c_\alpha^\dagger\right) + \frac{1}{4}\sum_{\alpha\beta\gamma}\left(V_{\beta\gamma}^{\gamma\alpha} - V_{\gamma\beta}^{\gamma\alpha}\right)\left(c_\alpha^\dagger c_\beta - c_\beta c_\alpha^\dagger\right)
$$

C. Reduced density matrix equations of motion

We start by solving for the normal rDM equation of motions

$$
\frac{d}{dt}\rho_{ab}^{\text{ee}}(t) = \left\langle \frac{dc_b^\dagger(t)}{dt} c_a(t)\right\rangle + \left\langle c_b^\dagger(t)\frac{dc_a(t)}{dt}\right\rangle, \tag{13}
$$

in the Heisenberg picture of quantum mechanics, where the fermionic operators evolve accordingly to the Heisenberg equation. Therefore the rDM equation of motion reads

$$
\frac{d}{dt}\rho_{ab}^{\text{ee}} = \left\langle \frac{i}{\hbar}[H, c_b^\dagger] c_a\right\rangle + \left\langle c_b^\dagger \frac{i}{\hbar}[H, c_a]\right\rangle. \tag{14}
$$

where we have omitted the time dependency in the fermionic operators for compactness. This practice is done throughout the remaining work with no further mention. As shown in Appendix, given the equal-time fermionic anti-commutator properties $\left\{c_a, c_b^\dagger\right\} = \left\{c_b^\dagger, c_a\right\} = \delta_{ab}$ and $\{c_a, c_b\} = \left\{c_b^\dagger, c_a^\dagger\right\} = 0$, each of the commutators are evaluated to

$$
[H, c_b^\dagger] = \bar{h}_{\alpha b} c_\alpha^\dagger + \frac{1}{4}\left(V_{\gamma b}^{\alpha\beta} - V_{b\gamma}^{\alpha\beta}\right)\left(c_\alpha^\dagger c_\beta^\dagger c_\gamma - c_\gamma c_\beta^\dagger c_\alpha^\dagger\right) \tag{15}
$$

$$
[H, c_a] = -\bar{h}_{a\alpha} c_\alpha + \frac{1}{4}\left(V_{\beta\gamma}^{\alpha a} - V_{\beta\gamma}^{a\alpha}\right)\left(c_\alpha^\dagger c_\beta c_\gamma - c_\gamma c_\beta c_\alpha^\dagger\right), \tag{16}
$$

where we have hidden the summations on repeated indices using Einstein's notation.

Commutators calculations

First we calculate the commutator with $c_b^\dagger$ making use of the following commutator properties. We obtain

$$\left[c_\alpha^\dagger c_\beta, c_b^\dagger\right] = c_\alpha^\dagger \left\{c_\beta, c_b^\dagger\right\} - \cancel{\left\{c_b^\dagger, c_\alpha^\dagger\right\}} c_\beta = c_\alpha^\dagger \delta_{\beta b}$$
$$\left[c_\beta c_\alpha^\dagger, c_b^\dagger\right] = c_\beta \cancel{\left\{c_\alpha^\dagger, c_b^\dagger\right\}} - \left\{c_b^\dagger, c_\beta\right\} c_\alpha^\dagger = -c_\alpha^\dagger \delta_{\beta b}$$

and that

$$\left[c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta, c_b^\dagger\right] = c_\alpha^\dagger c_\beta^\dagger \left(c_\gamma \left\{c_\delta, c_b^\dagger\right\} - \left\{c_b^\dagger, c_\gamma\right\} c_\delta\right) + \left(c_\alpha^\dagger \cancel{\left\{c_\beta^\dagger, c_b^\dagger\right\}} - \cancel{\left\{c_b^\dagger, c_\alpha^\dagger\right\}} c_\beta^\dagger\right) c_\gamma c_\delta = c_\alpha^\dagger c_\beta^\dagger c_\gamma \delta_{\delta b} - c_\alpha^\dagger c_\beta^\dagger c_\delta \delta_{\gamma b}$$
$$\left[c_\delta c_\gamma c_\beta^\dagger c_\alpha^\dagger, c_b^\dagger\right] = c_\delta c_\gamma \left(c_\beta^\dagger \cancel{\left\{c_\alpha^\dagger, c_b^\dagger\right\}} - \cancel{\left\{c_b^\dagger, c_\beta^\dagger\right\}} c_\alpha^\dagger\right) + \left(c_\delta \left\{c_\gamma, c_b^\dagger\right\} - \left\{c_b^\dagger, c_\delta\right\} c_\gamma\right) c_\beta^\dagger c_\alpha^\dagger = c_\delta c_\beta^\dagger c_\alpha^\dagger \delta_{\gamma b} - c_\gamma c_\beta^\dagger c_\alpha^\dagger \delta_{\delta b}$$

The complete commutator then reads

$$\begin{aligned}
[H, c_b^\dagger] &= \left[\frac{1}{2}\bar{h}_{\alpha\beta}\left(c_\alpha^\dagger c_\beta - c_\beta c_\alpha^\dagger\right) + \frac{1}{4}V_{\gamma\delta}^{\alpha\beta}\left(c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta + c_\delta c_\gamma c_\beta^\dagger c_\alpha^\dagger\right) + \frac{1}{2}\cancel{\tilde{h}_0}, c_b^\dagger\right] \\
&= \frac{1}{2}\bar{h}_{\alpha\beta}\left(\left[c_\alpha^\dagger c_\beta, c_b^\dagger\right] - \left[c_\beta c_\alpha^\dagger, c_b^\dagger\right]\right) + \frac{1}{4}V_{\gamma\delta}^{\alpha\beta}\left(\left[c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta, c_b^\dagger\right] + \left[c_\delta c_\gamma c_\beta^\dagger c_\alpha^\dagger, c_b^\dagger\right]\right) \\
&= \bar{h}_{\alpha\beta} c_\alpha^\dagger \delta_{\beta b} + \frac{1}{4}\left(V_{\gamma\delta}^{\alpha\beta} c_\alpha^\dagger c_\beta^\dagger c_\gamma \delta_{\delta b} - V_{\gamma\delta}^{\alpha\beta} c_\alpha^\dagger c_\beta^\dagger c_\delta \delta_{\gamma b} + V_{\gamma\delta}^{\alpha\beta} c_\delta c_\beta^\dagger c_\alpha^\dagger \delta_{\gamma b} - V_{\gamma\delta}^{\alpha\beta} c_\gamma c_\beta^\dagger c_\alpha^\dagger \delta_{\delta b}\right) \\
&= \bar{h}_{\alpha b} c_\alpha^\dagger + \frac{1}{4}\left(V_{\gamma b}^{\alpha\beta} - V_{b\gamma}^{\alpha\beta}\right)\left(c_\alpha^\dagger c_\beta^\dagger c_\gamma - c_\gamma c_\beta^\dagger c_\alpha^\dagger\right)
\end{aligned}$$

Analogous for the $[H, c_a]$ commutator, we have

$$\left[c_\alpha^\dagger c_\beta, c_a\right] = c_\alpha^\dagger \cancel{\{c_\beta, c_a\}} - \left\{c_a, c_\alpha^\dagger\right\} c_\beta = -c_\beta \delta_{a\alpha}$$
$$\left[c_\beta c_\alpha^\dagger, c_a\right] = c_\beta \left\{c_\alpha^\dagger, c_a\right\} - \cancel{\{c_a, c_\beta\}} c_\alpha^\dagger = c_\beta \delta_{a\alpha}$$

and that

$$\left[c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta, c_a\right] = c_\alpha^\dagger c_\beta^\dagger \left(c_\gamma \cancel{\{c_\delta, c_a\}} - \cancel{\{c_a, c_\gamma\}} c_\delta\right) + \left(c_\alpha^\dagger \left\{c_\beta^\dagger, c_a\right\} - \left\{c_a, c_\alpha^\dagger\right\} c_\beta^\dagger\right) c_\gamma c_\delta = c_\alpha^\dagger c_\gamma c_\delta \delta_{\beta a} - c_\beta^\dagger c_\gamma c_\delta \delta_{a\alpha}$$
$$\left[c_\delta c_\gamma c_\beta^\dagger c_\alpha^\dagger, c_a\right] = c_\delta c_\gamma \left(c_\beta^\dagger \left\{c_\alpha^\dagger, c_a\right\} - \left\{c_a, c_\beta^\dagger\right\} c_\alpha^\dagger\right) + \left(c_\delta \cancel{\{c_\gamma, c_a\}} - \cancel{\{c_a, c_\delta\}} c_\gamma\right) c_\beta^\dagger c_\alpha^\dagger = c_\delta c_\gamma c_\beta^\dagger \delta_{\alpha a} - c_\delta c_\gamma c_\alpha^\dagger \delta_{a\beta}$$

The complete commutator then reads

$$\begin{aligned}
[H, c_a] &= \left[\frac{1}{2}\bar{h}_{\alpha\beta}\left(c_\alpha^\dagger c_\beta - c_\beta c_\alpha^\dagger\right) + \frac{1}{4}V_{\gamma\delta}^{\alpha\beta}\left(c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta + c_\delta c_\gamma c_\beta^\dagger c_\alpha^\dagger\right) + \frac{1}{2}\cancel{\tilde{h}_0}, c_a\right] \\
&= \frac{1}{2}\bar{h}_{\alpha\beta}\left(\left[c_\alpha^\dagger c_\beta, c_a\right] - \left[c_\beta c_\alpha^\dagger, c_a\right]\right) + \frac{1}{4}V_{\gamma\delta}^{\alpha\beta}\left(\left[c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta, c_a\right] + \left[c_\delta c_\gamma c_\beta^\dagger c_\alpha^\dagger, c_a\right]\right) \\
&= -\bar{h}_{\alpha\beta} c_\beta \delta_{a\alpha} + \frac{1}{4}\left(V_{\gamma\delta}^{\alpha\beta} c_\alpha^\dagger c_\gamma c_\delta \delta_{\beta a} - V_{\gamma\delta}^{\alpha\beta} c_\beta^\dagger c_\gamma c_\delta \delta_{a\alpha} + V_{\gamma\delta}^{\alpha\beta} c_\delta c_\gamma c_\beta^\dagger \delta_{\alpha a} - V_{\gamma\delta}^{\alpha\beta} c_\delta c_\gamma c_\alpha^\dagger \delta_{a\beta}\right) \\
&= -\bar{h}_{a\beta} c_\beta + \frac{1}{4}\left(V_{\gamma\delta}^{\alpha a} c_\alpha^\dagger c_\gamma c_\delta - V_{\gamma\delta}^{a\beta} c_\beta^\dagger c_\gamma c_\delta + V_{\gamma\delta}^{a\beta} c_\delta c_\gamma c_\beta^\dagger - V_{\gamma\delta}^{\alpha a} c_\delta c_\gamma c_\alpha^\dagger\right) \\
&= -\bar{h}_{a\alpha} c_\alpha + \frac{1}{4}\left(V_{\beta\gamma}^{\alpha a} - V_{\beta\gamma}^{a\alpha}\right)\left(c_\alpha^\dagger c_\beta c_\gamma - c_\gamma c_\beta c_\alpha^\dagger\right)
\end{aligned}$$

$\{c_a, c_b^\dagger\} = \{c_b^\dagger, c_a\} = \delta_{ab}$
$\{c_a, c_b\} = \{c_b^\dagger, c_a^\dagger\} = 0$

$[AB, Z] = A\{B, Z\} - \{Z, A\} B$
$[AB, Z] = A[B, Z] - [Z, A]B$
$[AB, CD] = -A\left(C\{D, B\} - \{B, C\} D\right) - \left(C\{D, A\} - \{A, C\} D\right) B$
$[ABCD, Z] = AB\left(C\{D, Z\} - \{Z, C\} D\right) + \left(A\{B, Z\} - \{Z, A\} B\right) CD$

Substituting Eqs.(15) and (16) back into Eq.(14) yields

$$\begin{aligned}-i\hbar\frac{d}{dt}\rho^{\mathrm{ee}}_{ab} &= \rho^{\mathrm{ee}}_{a\alpha}\bar{h}_{\alpha b} - \bar{h}_{a\alpha}\rho^{\mathrm{ee}}_{\alpha b} \\ &+ \frac{1}{4}\left(V^{\alpha\beta}_{\gamma b} - V^{\alpha\beta}_{b\gamma}\right)\left(\left\langle c^\dagger_\alpha c^\dagger_\beta c_\gamma c_a\right\rangle - \left\langle c_\gamma c^\dagger_\beta c^\dagger_\alpha c_a\right\rangle\right) \\ &+ \frac{1}{4}\left(V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma}\right)\left(\left\langle c^\dagger_b c^\dagger_\alpha c_\beta c_\gamma\right\rangle - \left\langle c^\dagger_b c_\gamma c_\beta c^\dagger_\alpha\right\rangle\right)\end{aligned} \tag{17}$$

where we came back to the definition of the normal rDM in Eq.(4) for the first two terms. Notice, however, that the remaining interaction terms correspond to the expectation values of four-operators. For this, we introduce a mean-field approximation where we assume the two-particle expectation value to simply behave as a product of two one-particle expectation values. From this mean-field decoupling we can then make use of *Wick's theorem* as done in the Appendix. We obtain

$$\left\langle c^\dagger_\alpha c^\dagger_\beta c_\gamma c_\delta\right\rangle \approx \rho^{\mathrm{he}}_{\beta\alpha}\rho^{\mathrm{eh}}_{\delta\gamma} - \rho^{\mathrm{ee}}_{\gamma\alpha}\rho^{\mathrm{hh}}_{\delta\beta} + \rho^{\mathrm{ee}}_{\delta\alpha}\rho^{\mathrm{ee}}_{\gamma\beta} \tag{18}$$

$$\left\langle c_\alpha c^\dagger_\beta c^\dagger_\gamma c_\delta\right\rangle \approx \rho^{\mathrm{hh}}_{\beta\alpha}\rho^{\mathrm{ee}}_{\delta\gamma} - \rho^{\mathrm{hh}}_{\gamma\alpha}\rho^{\mathrm{ee}}_{\delta\beta} + \rho^{\mathrm{eh}}_{\delta\alpha}\rho^{\mathrm{he}}_{\gamma\beta} \tag{19}$$

$$\left\langle c^\dagger_\alpha c_\beta c_\gamma c^\dagger_\delta\right\rangle \approx \rho^{\mathrm{ee}}_{\beta\alpha}\rho^{\mathrm{ee}}_{\delta\gamma} - \rho^{\mathrm{ee}}_{\gamma\alpha}\rho^{\mathrm{hh}}_{\delta\beta} + \rho^{\mathrm{he}}_{\delta\alpha}\rho^{\mathrm{eh}}_{\gamma\beta} \tag{20}$$

As derived in details in the Appendix, the rDM equation of motion can then be written in a time-dependent Hartree-Fock in a more compact and elegant manner as

$$i\hbar\frac{d}{dt}\boldsymbol{\rho}_{\mathrm{ee}} = [\mathbf{H}_{\mathrm{HF}}, \boldsymbol{\rho}_{\mathrm{ee}}] - \boldsymbol{\rho}_{\mathrm{eh}}\boldsymbol{\Delta}^\dagger + \boldsymbol{\Delta}\boldsymbol{\rho}_{\mathrm{he}} \tag{21}$$

where, for the normal terms, we defined $\mathbf{H}_{\mathrm{HF}}$ as the Hartree-Fock Hamiltonian

$$\mathbf{H}_{HF} \equiv \boldsymbol{h} + \boldsymbol{B}\cdot\mathbf{F}(t) + \boldsymbol{\Sigma}^H + \boldsymbol{\Sigma}^F \tag{22}$$

identifying the Hartree $\boldsymbol{\Sigma}^H$and Fock $\boldsymbol{\Sigma}^F$ self-energy terms respectively as

$$\Sigma^H_{ab} \equiv \Sigma^H_{ab}[\boldsymbol{\rho}_{\mathrm{ee}}(t)] = V^{a\delta}_{\gamma b}\rho^{\mathrm{ee}}_{\gamma\delta}, \tag{23}$$

$$\Sigma^F_{ab} \equiv \Sigma^F_{ab}[\boldsymbol{\rho}_{\mathrm{ee}}(t)] = -V^{a\delta}_{b\gamma}\rho^{\mathrm{ee}}_{\gamma\delta}, \tag{24}$$

and the anomalous self-energies, also known as pairing, as

$$\Delta_{ab} \equiv \Delta_{ab}[\boldsymbol{\rho}_{\mathrm{eh}}(t)] \equiv V^{ab}_{\delta\gamma}\rho^{\mathrm{eh}}_{\gamma\delta} \tag{25}$$

Notice that the Fock self-energy term in Eq.(24) is an attractive interaction (due the negative sign) such that it lower the total energy of the system enabling the formation of bound electron-hole states. Intuitively, the source of this attractive term can be thought as due to the opposing charges of the electron and hole. Conversely, the Hartree self-energy term will control details of the excitation spectrum, such as spin-splitting (which will be discussed further in the text). As for an intuitive picture for the anomalous self-energy term, one think of it in terms of BCS superconductivity; as an electron propagates through an overall neutral lattice, the positive-ions are slightly displaced creating a positively-charged region which will in turn attract the other electron in the pair, creating an effective attractive interaction between them. Meh, saying it's attractive induces one to define it with a minus sign which is wrong for the general case. Is it though? I'm just that I don't want the minus there because then, in the final result, I end up with a wrong minus sign.

Normal rDM EoM calculations
For the non-interaction terms we simply have

$$\rho^{\mathrm{ee}}_{a\alpha}\bar{h}_{\alpha b} - \bar{h}_{a\alpha}\rho^{\mathrm{ee}}_{\alpha b} = \boldsymbol{\rho}_{\mathrm{ee}}\bar{\boldsymbol{h}} - \bar{\boldsymbol{h}}\boldsymbol{\rho}_{\mathrm{ee}} = -\left[\bar{\boldsymbol{h}}, \boldsymbol{\rho}_{\mathrm{ee}}\right]$$

For the interaction terms in Eq.(17), we perform a mean-field decoupling on the 4-operator terms, reading

$$\begin{aligned}&\frac{1}{4}\left(V^{\alpha\beta}_{\gamma b}-V^{\alpha\beta}_{b\gamma}\right)\left(\left[\rho^{\mathrm{he}}_{\beta\alpha}\rho^{\mathrm{eh}}_{a\gamma}-\rho^{\mathrm{ee}}_{\gamma\alpha}\rho^{\mathrm{ee}}_{a\beta}+\rho^{\mathrm{ee}}_{a\alpha}\rho^{\mathrm{ee}}_{\gamma\beta}\right]-\left[\rho^{\mathrm{hh}}_{\beta\gamma}\rho^{\mathrm{ee}}_{a\alpha}-\rho^{\mathrm{hh}}_{\alpha\gamma}\rho^{\mathrm{ee}}_{a\beta}+\rho^{\mathrm{eh}}_{a\gamma}\rho^{\mathrm{he}}_{\alpha\beta}\right]\right)\\&+\frac{1}{4}\left(V^{\alpha a}_{\beta\gamma}-V^{a\alpha}_{\beta\gamma}\right)\left(\left[\rho^{\mathrm{he}}_{\alpha b}\rho^{\mathrm{eh}}_{\gamma\beta}-\rho^{\mathrm{ee}}_{\beta b}\rho^{\mathrm{ee}}_{\gamma\alpha}+\rho^{\mathrm{ee}}_{\gamma b}\rho^{\mathrm{ee}}_{\beta\alpha}\right]-\left[\rho^{\mathrm{ee}}_{\gamma b}\rho^{\mathrm{hh}}_{\alpha\beta}-\rho^{\mathrm{ee}}_{\beta b}\rho^{\mathrm{hh}}_{\alpha\gamma}+\rho^{\mathrm{he}}_{\alpha b}\rho^{\mathrm{eh}}_{\beta\gamma}\right]\right)\end{aligned}$$

Focusing only on the terms with normal rDM, we have

$$\begin{aligned}&\frac{1}{4}\left(V^{\alpha\beta}_{\gamma b}-V^{\alpha\beta}_{b\gamma}\right)\left[\rho^{\mathrm{ee}}_{a\beta}\left(-\rho^{\mathrm{ee}}_{\gamma\alpha}+\rho^{\mathrm{hh}}_{\alpha\gamma}\right)+\rho^{\mathrm{ee}}_{a\alpha}\left(\rho^{\mathrm{ee}}_{\gamma\beta}-\rho^{\mathrm{hh}}_{\beta\gamma}\right)\right]+\frac{1}{4}\left(V^{\alpha a}_{\beta\gamma}-V^{a\alpha}_{\beta\gamma}\right)\left[\rho^{\mathrm{ee}}_{\beta b}\left(-\rho^{\mathrm{ee}}_{\gamma\alpha}+\rho^{\mathrm{hh}}_{\alpha\gamma}\right)+\rho^{\mathrm{ee}}_{\gamma b}\left(\rho^{\mathrm{ee}}_{\beta\alpha}-\rho^{\mathrm{hh}}_{\alpha\beta}\right)\right]\\=&\frac{1}{4}\rho^{\mathrm{ee}}_{a\alpha}\left[-\left(V^{\beta\alpha}_{\gamma b}-V^{\beta\alpha}_{b\gamma}\right)+\left(V^{\alpha\beta}_{\gamma b}-V^{\alpha\beta}_{b\gamma}\right)\right]\left(\rho^{\mathrm{ee}}_{\gamma\beta}-\rho^{\mathrm{hh}}_{\beta\gamma}\right)+\frac{1}{4}\left[-\left(V^{\alpha a}_{\beta\gamma}-V^{a\alpha}_{\beta\gamma}\right)+\left(V^{\alpha a}_{\gamma\beta}-V^{a\alpha}_{\gamma\beta}\right)\right]\left(\rho^{\mathrm{ee}}_{\gamma\alpha}-\rho^{\mathrm{hh}}_{\alpha\gamma}\right)\rho^{\mathrm{ee}}_{\beta b}\\=&\frac{1}{2}\rho^{\mathrm{ee}}_{a\alpha}\left[V^{\beta\alpha}_{b\gamma}-V^{\beta\alpha}_{\gamma b}\right]\left(\rho^{\mathrm{ee}}_{\gamma\beta}-\rho^{\mathrm{hh}}_{\beta\gamma}\right)+\frac{1}{2}\left[V^{a\alpha}_{\beta\gamma}-V^{\alpha a}_{\beta\gamma}\right]\left(\rho^{\mathrm{ee}}_{\gamma\alpha}-\rho^{\mathrm{hh}}_{\alpha\gamma}\right)\rho^{\mathrm{ee}}_{\beta b}\\=&\frac{1}{2}\rho^{\mathrm{ee}}_{a\alpha}\left[V^{\beta\alpha}_{b\gamma}-V^{\beta\alpha}_{\gamma b}\right]\left(2\rho^{\mathrm{ee}}_{\gamma\beta}-\delta_{\beta\gamma}\right)+\frac{1}{2}\left[V^{a\alpha}_{\beta\gamma}-V^{\alpha a}_{\beta\gamma}\right]\left(2\rho^{\mathrm{ee}}_{\gamma\alpha}-\delta_{\alpha\gamma}\right)\rho^{\mathrm{ee}}_{\beta b}\\=&\left(\rho^{\mathrm{ee}}_{a\alpha}\left[\Sigma^{H}_{\alpha b}+\Sigma^{F}_{\alpha b}\right]-\left[\Sigma^{H}_{a\beta}+\Sigma^{F}_{a\beta}\right]\rho^{\mathrm{ee}}_{\beta b}\right)-\left(\rho^{\mathrm{ee}}_{a\alpha}\mu_{\alpha b}-\mu_{a\beta}\rho^{\mathrm{ee}}_{\beta b}\right)\\=&\left[-\boldsymbol{\Sigma}+\boldsymbol{\mu},\boldsymbol{\rho}_{\mathrm{ee}}\right]\end{aligned}$$

The remaining terms, corresponding to the anomalous terms read

$$\begin{aligned}&\frac{1}{4}\left(V^{\alpha\beta}_{\gamma b}-V^{\alpha\beta}_{b\gamma}\right)\rho^{\mathrm{he}}_{\beta\alpha}\rho^{\mathrm{eh}}_{a\gamma}-\frac{1}{4}\left(V^{\alpha\beta}_{\gamma b}-V^{\alpha\beta}_{b\gamma}\right)\rho^{\mathrm{eh}}_{a\gamma}\rho^{\mathrm{he}}_{\alpha\beta}+\frac{1}{4}\left(V^{\alpha a}_{\beta\gamma}-V^{a\alpha}_{\beta\gamma}\right)\rho^{\mathrm{he}}_{\alpha b}\rho^{\mathrm{eh}}_{\gamma\beta}-\frac{1}{4}\left(V^{\alpha a}_{\beta\gamma}-V^{a\alpha}_{\beta\gamma}\right)\rho^{\mathrm{he}}_{\alpha b}\rho^{\mathrm{eh}}_{\beta\gamma}\\=&\frac{1}{4}\left(V^{\alpha\beta}_{\gamma b}-V^{\alpha\beta}_{b\gamma}-V^{\beta\alpha}_{\gamma b}+V^{\beta\alpha}_{b\gamma}\right)\rho^{\mathrm{he}}_{\beta\alpha}\rho^{\mathrm{eh}}_{a\gamma}+\frac{1}{4}\left(V^{\alpha a}_{\beta\gamma}-V^{a\alpha}_{\beta\gamma}-V^{\alpha a}_{\gamma\beta}+V^{a\alpha}_{\gamma\beta}\right)\rho^{\mathrm{he}}_{\alpha b}\rho^{\mathrm{eh}}_{\gamma\beta}\\=&\frac{1}{2}\rho^{\mathrm{eh}}_{a\gamma}\left[-\left(V^{b\gamma}_{\beta\alpha}-V^{\gamma b}_{\beta\alpha}\right)\rho^{\mathrm{eh}}_{\alpha\beta}\right]^{\dagger}-\frac{1}{2}\left(V^{a\alpha}_{\beta\gamma}-V^{\alpha a}_{\beta\gamma}\right)\rho^{\mathrm{eh}}_{\gamma\beta}\rho^{\mathrm{he}}_{\alpha b}\\=&-\frac{1}{2}\rho^{\mathrm{eh}}_{a\gamma}\left(\Delta_{b\gamma}-\Delta_{\gamma b}\right)^{\dagger}-\frac{1}{2}\left(\Delta_{a\alpha}-\Delta_{\alpha a}\right)\rho^{\mathrm{he}}_{\alpha b}\\=&\rho^{\mathrm{eh}}_{a\gamma}\Delta^{\dagger}_{\gamma b}-\Delta_{a\alpha}\rho^{\mathrm{he}}_{\alpha b}\\=&\boldsymbol{\rho}_{\mathrm{eh}}\boldsymbol{\Delta}^{\dagger}-\boldsymbol{\Delta}\boldsymbol{\rho}_{\mathrm{he}}\end{aligned}$$

Putting all the terms together and bewaring the minus signs in Eq.(14) left behind, yields

$$\begin{aligned}&-\left[\bar{\boldsymbol{h}},\boldsymbol{\rho}_{\mathrm{ee}}\right]+\left[-\boldsymbol{\Sigma}+\boldsymbol{\mu},\boldsymbol{\rho}_{\mathrm{ee}}\right]+\left[\boldsymbol{\rho}_{\mathrm{eh}}\boldsymbol{\Delta}^{\dagger}-\boldsymbol{\Delta}\boldsymbol{\rho}_{\mathrm{he}}\right]\\=&-\left[\left(\bar{\boldsymbol{h}}-\boldsymbol{\mu}\right)+\boldsymbol{\Sigma},\boldsymbol{\rho}_{\mathrm{ee}}\right]+\boldsymbol{\rho}_{\mathrm{he}}\boldsymbol{\Delta}^{\dagger}-\boldsymbol{\Delta}\boldsymbol{\rho}_{\mathrm{he}}\\=&-\left[\mathbf{H}_{\mathrm{HF}},\boldsymbol{\rho}_{\mathrm{ee}}\right]+\boldsymbol{\rho}_{\mathrm{eh}}\boldsymbol{\Delta}^{\dagger}-\boldsymbol{\Delta}\boldsymbol{\rho}_{\mathrm{he}}\end{aligned}$$

$$\begin{aligned}&\rho^{\mathrm{hh}}_{ab}=\left\langle c_b c^{\dagger}_a\right\rangle=-\left\langle c^{\dagger}_a c_b\right\rangle+\delta_{ab}=-\rho^{\mathrm{eh}}_{ba}+\delta_{ab}\\&\rho^{\mathrm{he}}_{ab}=\left\langle c^{\dagger}_b c^{\dagger}_a\right\rangle=\left\langle c_a c_b\right\rangle^{\dagger}=\left(\rho^{\mathrm{eh}}_{ba}\right)^{\dagger}\\&\left\{c_a,c_b\right\}=0\Leftrightarrow c_a c_b=-c_b c_a\Rightarrow\rho^{\mathrm{eh}}_{ba}=-\rho^{\mathrm{eh}}_{ab}\\&\left\{c^{\dagger}_a,c^{\dagger}_b\right\}=0\Leftrightarrow c^{\dagger}_a c^{\dagger}_b=-c^{\dagger}_b c^{\dagger}_a\Rightarrow\rho^{\mathrm{eh}}_{ba}=-\rho^{\mathrm{eh}}_{ab}\\&\Delta_{ab}=V^{ab}_{\delta\gamma}\rho^{\mathrm{eh}}_{\gamma\delta}=V^{ba}_{\gamma\delta}\rho^{\mathrm{eh}}_{\gamma\delta}\overset{\delta\rightleftarrows\gamma}{=}V^{ba}_{\delta\gamma}\rho^{\mathrm{eh}}_{\delta\gamma}=V^{ba}_{\delta\gamma}\left(-\rho^{\mathrm{eh}}_{\gamma\delta}\right)=-\Delta_{ba}\end{aligned}$$

With the normal EoM expressed, we now focus on the anomalous EoM,

$$\frac{d}{dt}\rho^{\mathrm{eh}}_{ab}=\left\langle\frac{i}{\hbar}[H,c_b]c_a\right\rangle-\left\langle c_b\frac{i}{\hbar}[H,c_a]\right\rangle. \tag{26}$$

where we instead perform the mean-field decoupling

$$\left\langle c_\alpha^\dagger c_\beta c_\gamma c_\delta \right\rangle \approx \rho^{\mathrm{ee}}_{\beta\alpha}\rho^{\mathrm{eh}}_{\delta\gamma} - \rho^{\mathrm{eh}}_{\gamma\alpha}\rho^{\mathrm{eh}}_{\delta\beta} + \rho^{\mathrm{ee}}_{\delta\alpha}\rho^{\mathrm{eh}}_{\gamma\beta}, \tag{27}$$

$$\left\langle c_\alpha c_\beta^\dagger c_\gamma c_\delta \right\rangle \approx \rho^{\mathrm{he}}_{\beta\alpha}\rho^{\mathrm{eh}}_{\delta\gamma} - \rho^{\mathrm{eh}}_{\gamma\alpha}\rho^{\mathrm{ee}}_{\delta\beta} + \rho^{\mathrm{eh}}_{\delta\alpha}\rho^{\mathrm{ee}}_{\gamma\beta}. \tag{28}$$

$$\left\langle c_\alpha c_\beta c_\gamma^\dagger c_\delta \right\rangle \approx \rho^{\mathrm{eh}}_{\beta\alpha}\rho^{\mathrm{ee}}_{\delta\gamma} - \rho^{\mathrm{hh}}_{\gamma\alpha}\rho^{\mathrm{eh}}_{\delta\beta} + \rho^{\mathrm{eh}}_{\delta\alpha}\rho^{\mathrm{hh}}_{\gamma\beta} \tag{29}$$

$$\left\langle c_\alpha c_\beta c_\gamma c_\delta^\dagger \right\rangle \approx \rho^{\mathrm{eh}}_{\beta\alpha}\rho^{\mathrm{hh}}_{\delta\gamma} - \rho^{\mathrm{eh}}_{\gamma\alpha}\rho^{\mathrm{hh}}_{\delta\beta} + \rho^{\mathrm{hh}}_{\delta\alpha}\rho^{\mathrm{eh}}_{\gamma\beta} \tag{30}$$

This yields the anomalous equation of motion

$$i\hbar\frac{d}{dt}\boldsymbol{\rho}_{\mathrm{eh}} = \mathbf{H}_{\mathrm{HF}}\boldsymbol{\rho}_{\mathrm{eh}} + \boldsymbol{\rho}_{\mathrm{eh}}\mathbf{H}_{\mathrm{HF}}^T + \boldsymbol{\Delta}\boldsymbol{\rho}_{\mathrm{hh}} - \boldsymbol{\rho}_{\mathrm{ee}}\boldsymbol{\Delta} \tag{31}$$

Anomalous rDM EoM calculations
Using the result shown in Eq.(16), the EoM in Eq.(26) reads

$$\begin{aligned} i\hbar\frac{d}{dt}\rho^{\mathrm{eh}}_{ab} =& -\bar{h}_{b\alpha}\rho^{\mathrm{eh}}_{a\alpha} + \frac{1}{4}\left(V^{\alpha b}_{\beta\gamma} - V^{b\alpha}_{\beta\gamma}\right)\left(\left\langle c_\alpha^\dagger c_\beta c_\gamma c_a\right\rangle - \left\langle c_\gamma c_\beta c_\alpha^\dagger c_a\right\rangle\right) \\ & -\bar{h}_{a\alpha}\rho^{\mathrm{eh}}_{\alpha b} + \frac{1}{4}\left(V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma}\right)\left(\left\langle c_b c_\alpha^\dagger c_\beta c_\gamma\right\rangle - \left\langle c_b c_\gamma c_\beta c_\alpha^\dagger\right\rangle\right) \end{aligned}$$

Immediately, the single particle term reads

$$-\bar{h}_{b\alpha}\rho^{\mathrm{eh}}_{a\alpha} - \bar{h}_{a\alpha}\rho^{\mathrm{eh}}_{\alpha b} = -\boldsymbol{\rho}_{\mathrm{eh}}\bar{\boldsymbol{h}}^T - \bar{\boldsymbol{h}}\boldsymbol{\rho}_{\mathrm{eh}}$$

From Wick's theorem we have that

$$\begin{aligned} &\frac{1}{4}\left(V^{\alpha b}_{\beta\gamma} - V^{b\alpha}_{\beta\gamma}\right)\left(\left[\rho^{\mathrm{ee}}_{\beta\alpha}\rho^{\mathrm{eh}}_{a\gamma} - \rho^{\mathrm{ee}}_{\gamma\alpha}\rho^{\mathrm{eh}}_{a\beta} + \rho^{\mathrm{ee}}_{a\alpha}\rho^{\mathrm{eh}}_{\gamma\beta}\right] - \left[\rho^{\mathrm{eh}}_{\beta\gamma}\rho^{\mathrm{ee}}_{a\alpha} - \rho^{\mathrm{hh}}_{\alpha\gamma}\rho^{\mathrm{eh}}_{a\beta} + \rho^{\mathrm{eh}}_{a\gamma}\rho^{\mathrm{hh}}_{\alpha\beta}\right]\right) \\ +&\frac{1}{4}\left(V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma}\right)\left(\left[\rho^{\mathrm{hh}}_{\alpha b}\rho^{\mathrm{eh}}_{\gamma\beta} - \rho^{\mathrm{eh}}_{\beta b}\rho^{\mathrm{ee}}_{\gamma\alpha} + \rho^{\mathrm{eh}}_{\gamma b}\rho^{\mathrm{ee}}_{\beta\alpha}\right] - \left[\rho^{\mathrm{eh}}_{\gamma b}\rho^{\mathrm{hh}}_{\alpha\beta} - \rho^{\mathrm{eh}}_{\beta b}\rho^{\mathrm{hh}}_{\alpha\gamma} + \rho^{\mathrm{hh}}_{\alpha b}\rho^{\mathrm{eh}}_{\beta\gamma}\right]\right) \end{aligned}$$

Let us start by focusing only on the first two terms,

$$\begin{aligned} =&\frac{1}{4}\left(V^{\alpha b}_{\beta\gamma} - V^{b\alpha}_{\beta\gamma}\right)\rho^{\mathrm{ee}}_{\beta\alpha}\rho^{\mathrm{eh}}_{a\gamma} - \frac{1}{4}\left(V^{\alpha b}_{\beta\gamma} - V^{b\alpha}_{\beta\gamma}\right)\rho^{\mathrm{ee}}_{\gamma\alpha}\rho^{\mathrm{eh}}_{a\beta} + \frac{1}{4}\left(V^{\alpha b}_{\beta\gamma} - V^{b\alpha}_{\beta\gamma}\right)\rho^{\mathrm{ee}}_{a\alpha}\rho^{\mathrm{eh}}_{\gamma\beta} \\ &-\frac{1}{4}\left(V^{\alpha b}_{\beta\gamma} - V^{b\alpha}_{\beta\gamma}\right)\rho^{\mathrm{eh}}_{\beta\gamma}\rho^{\mathrm{ee}}_{a\alpha} + \frac{1}{4}\left(V^{\alpha b}_{\beta\gamma} - V^{b\alpha}_{\beta\gamma}\right)\rho^{\mathrm{hh}}_{\alpha\gamma}\rho^{\mathrm{eh}}_{a\beta} - \frac{1}{4}\left(V^{\alpha b}_{\beta\gamma} - V^{b\alpha}_{\beta\gamma}\right)\rho^{\mathrm{eh}}_{a\gamma}\rho^{\mathrm{hh}}_{\alpha\beta} \\ =&\frac{1}{2}\rho^{\mathrm{eh}}_{a\gamma}\left[\left(-V^{\alpha b}_{\gamma\beta} + V^{b\alpha}_{\gamma\beta}\right)\rho^{\mathrm{ee}}_{\beta\alpha} + \left(V^{\alpha b}_{\gamma\beta} - V^{b\alpha}_{\gamma\beta}\right)\left(-\rho^{\mathrm{ee}}_{\beta\alpha} + \delta_{\alpha\beta}\right)\right] - \frac{1}{2}\rho^{\mathrm{ee}}_{a\alpha}\left(V^{\alpha b}_{\beta\gamma} - V^{b\alpha}_{\beta\gamma}\right)\rho^{\mathrm{eh}}_{\beta\gamma} \\ =&\rho^{\mathrm{eh}}_{a\gamma}\left(-\Sigma^H_{b\gamma} - \Sigma^F_{b\gamma}\right) + \rho^{\mathrm{eh}}_{a\gamma}\mu_{b\gamma} - \frac{1}{2}\rho^{\mathrm{ee}}_{a\alpha}\left(\Delta_{b\alpha} - \Delta_{\alpha b}\right) \\ =&\boldsymbol{\rho}_{\mathrm{eh}}\left(-\boldsymbol{\Sigma}^T + \boldsymbol{\mu}^T\right) + \boldsymbol{\rho}_{\mathrm{ee}}\boldsymbol{\Delta} \end{aligned}$$

and next on the terms two terms

$$\begin{aligned} &\frac{1}{4}\left(V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma}\right)\rho^{\mathrm{hh}}_{\alpha b}\rho^{\mathrm{eh}}_{\gamma\beta} - \frac{1}{4}\left(V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma}\right)\rho^{\mathrm{eh}}_{\beta b}\rho^{\mathrm{ee}}_{\gamma\alpha} + \frac{1}{4}\left(V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma}\right)\rho^{\mathrm{eh}}_{\gamma b}\rho^{\mathrm{ee}}_{\beta\alpha} \\ &-\frac{1}{4}\left(V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma}\right)\rho^{\mathrm{eh}}_{\gamma b}\rho^{\mathrm{hh}}_{\alpha\beta} + \frac{1}{4}\left(V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma}\right)\rho^{\mathrm{eh}}_{\beta b}\rho^{\mathrm{hh}}_{\alpha\gamma} - \frac{1}{4}\left(V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma}\right)\rho^{\mathrm{hh}}_{\alpha b}\rho^{\mathrm{eh}}_{\beta\gamma} \\ =&\frac{1}{2}\left(V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma}\right)\rho^{\mathrm{eh}}_{\gamma\beta}\rho^{\mathrm{hh}}_{\alpha b} + \frac{1}{2}\left(-V^{\alpha a}_{\beta\gamma} + V^{a\alpha}_{\beta\gamma}\right)\left(-\rho^{\mathrm{ee}}_{\beta\alpha} + \delta_{\alpha\beta}\right)\rho^{\mathrm{eh}}_{\gamma b} + \frac{1}{2}\left(-V^{\alpha a}_{\gamma\beta} + V^{a\alpha}_{\gamma\beta}\right)\rho^{\mathrm{ee}}_{\beta\alpha}\rho^{\mathrm{eh}}_{\gamma b} \\ =&\frac{1}{2}\left(\Delta_{\alpha a} - \Delta_{a\alpha}\right)\rho^{\mathrm{hh}}_{\alpha b} + \frac{1}{2}\left(-\Sigma^F_{a\gamma} - \Sigma^H_{a\gamma}\right)\rho^{\mathrm{eh}}_{\gamma b} + \mu_{a\gamma}\rho^{\mathrm{eh}}_{\gamma b} + \frac{1}{2}\left(-\Sigma^H_{a\gamma} - \Sigma^F_{a\gamma}\right)\rho^{\mathrm{eh}}_{\gamma b} \\ =& -\Delta_{a\alpha}\rho^{\mathrm{hh}}_{\alpha b} - \Sigma_{a\gamma}\rho^{\mathrm{eh}}_{\gamma b} + \mu_{a\gamma}\rho^{\mathrm{eh}}_{\gamma b} \\ =&\left(-\boldsymbol{\Sigma} + \boldsymbol{\mu}\right)\boldsymbol{\rho}_{\mathrm{eh}} - \boldsymbol{\Delta}\boldsymbol{\rho}_{\mathrm{hh}} \end{aligned}$$

Putting these three terms together we obtain

$$\begin{aligned}
&\left[\boldsymbol{\rho}_{\mathrm{eh}}\left(-\boldsymbol{\Sigma}^T+\boldsymbol{\mu}^T\right)+\boldsymbol{\rho}_{\mathrm{ee}}\boldsymbol{\Delta}\right]+\left[\left(-\boldsymbol{\Sigma}+\boldsymbol{\mu}\right)\boldsymbol{\rho}_{\mathrm{eh}}-\boldsymbol{\Delta}\boldsymbol{\rho}_{\mathrm{hh}}\right]+\left[-\boldsymbol{\rho}_{\mathrm{eh}}\bar{\boldsymbol{h}}^T-\bar{\boldsymbol{h}}\boldsymbol{\rho}_{\mathrm{eh}}\right]\\
=&-\left(\left(\bar{\boldsymbol{h}}-\boldsymbol{\mu}\right)+\boldsymbol{\Sigma}\right)\boldsymbol{\rho}_{\mathrm{eh}}-\boldsymbol{\rho}_{\mathrm{eh}}\left(\left(\bar{\boldsymbol{h}}-\boldsymbol{\mu}\right)+\boldsymbol{\Sigma}\right)^T-\boldsymbol{\Delta}\boldsymbol{\rho}_{\mathrm{hh}}+\boldsymbol{\rho}_{\mathrm{ee}}\boldsymbol{\Delta}\\
=&-\mathbf{H}_{\mathrm{HF}}\boldsymbol{\rho}_{\mathrm{eh}}-\boldsymbol{\rho}_{\mathrm{eh}}\mathbf{H}_{\mathrm{HF}}^T-\boldsymbol{\Delta}\boldsymbol{\rho}_{\mathrm{hh}}+\boldsymbol{\rho}_{\mathrm{ee}}\boldsymbol{\Delta}
\end{aligned}$$

Finally, the complete reduced density matrix equation of motions yields

$$i\hbar\frac{d}{dt}\boldsymbol{\rho}=\left[\mathbf{H}_{\mathrm{BdG}},\boldsymbol{\rho}\right] \tag{32}$$

where we rewrote this expression in terms of an effective Nambu Hamiltonian $\mathbf{H}_{\mathrm{BdG}}$ defined as

$$\mathbf{H}_{\mathrm{BdG}}=\begin{bmatrix}\mathbf{H}_{\mathrm{HF}} & \boldsymbol{\Delta}\\ \boldsymbol{\Delta}^\dagger & -\mathbf{H}_{\mathrm{HF}}^T\end{bmatrix} \tag{33}$$

Furthermore, using the relations initially shown in Eqs.(7), one can express both the EoMs in Eq.(21) and Eq.(31) only in terms of the electronic normal $\boldsymbol{\rho}_{\mathrm{n}}\equiv\boldsymbol{\rho}_{\mathrm{ee}}$ and anomalous $\boldsymbol{\rho}_{\mathrm{a}}\equiv\boldsymbol{\rho}_{\mathrm{eh}}$ rDMs (instead of the four distinct terms $\boldsymbol{\rho}_{\mathrm{ee}}$, $\boldsymbol{\rho}_{\mathrm{eh}}$, $\boldsymbol{\rho}_{\mathrm{he}}$ and $\boldsymbol{\rho}_{\mathrm{hh}}$). We write

$$\begin{aligned}
i\hbar\frac{d}{dt}\boldsymbol{\rho}_{\mathrm{n}}=&\left[\mathbf{H}_{\mathrm{HF}},\boldsymbol{\rho}_{\mathrm{n}}\right]-\left(\boldsymbol{\rho}_{\mathrm{a}}\boldsymbol{\Delta}^\dagger+\left(\boldsymbol{\rho}_{\mathrm{a}}\boldsymbol{\Delta}^\dagger\right)^\dagger\right)\\
i\hbar\frac{d}{dt}\boldsymbol{\rho}_{\mathrm{a}}=&\mathbf{H}_{\mathrm{HF}}\boldsymbol{\rho}_{\mathrm{a}}+\boldsymbol{\rho}_{\mathrm{a}}\mathbf{H}_{\mathrm{HF}}^T-\left(\boldsymbol{\Delta}\boldsymbol{\rho}_{\mathrm{n}}^\dagger+\boldsymbol{\rho}_{\mathrm{n}}\boldsymbol{\Delta}\right)+\boldsymbol{\Delta}
\end{aligned}$$

BdG Hamiltonian
Confirming that the result actually checks out,

$$\begin{aligned}
\left[\mathbf{H}_{\mathrm{BdG}},\boldsymbol{\rho}\right]=&\begin{bmatrix}\mathbf{H} & \boldsymbol{\Delta}\\ \boldsymbol{\Delta}^\dagger & -\mathbf{H}^T\end{bmatrix}\begin{bmatrix}\boldsymbol{\rho}_{\mathrm{ee}} & \boldsymbol{\rho}_{\mathrm{eh}}\\ \boldsymbol{\rho}_{\mathrm{he}} & \boldsymbol{\rho}_{\mathrm{hh}}\end{bmatrix}-\begin{bmatrix}\boldsymbol{\rho}_{\mathrm{ee}} & \boldsymbol{\rho}_{\mathrm{eh}}\\ \boldsymbol{\rho}_{\mathrm{he}} & \boldsymbol{\rho}_{\mathrm{hh}}\end{bmatrix}\begin{bmatrix}\mathbf{H} & \boldsymbol{\Delta}\\ \boldsymbol{\Delta}^\dagger & -\mathbf{H}^T\end{bmatrix}\\
=&\begin{bmatrix}\left(\mathbf{H}\boldsymbol{\rho}_{\mathrm{ee}}+\boldsymbol{\Delta}\boldsymbol{\rho}_{\mathrm{he}}\right) & \left(\mathbf{H}\boldsymbol{\rho}_{\mathrm{eh}}+\boldsymbol{\Delta}\boldsymbol{\rho}_{\mathrm{hh}}\right)\\ \circ & \circ\end{bmatrix}-\begin{bmatrix}\left(\boldsymbol{\rho}_{\mathrm{ee}}\mathbf{H}+\boldsymbol{\rho}_{\mathrm{eh}}\boldsymbol{\Delta}^\dagger\right) & \left(\boldsymbol{\rho}_{\mathrm{ee}}\boldsymbol{\Delta}-\boldsymbol{\rho}_{\mathrm{eh}}\mathbf{H}^T\right)\\ \circ & \circ\end{bmatrix}\\
=&\begin{bmatrix}\left(\left[\mathbf{H},\boldsymbol{\rho}_{\mathrm{ee}}\right]-\boldsymbol{\rho}_{\mathrm{eh}}\boldsymbol{\Delta}^\dagger+\boldsymbol{\Delta}\boldsymbol{\rho}_{\mathrm{he}}\right) & \left(\mathbf{H}\boldsymbol{\rho}_{\mathrm{eh}}+\boldsymbol{\rho}_{\mathrm{eh}}\mathbf{H}^T+\boldsymbol{\Delta}\boldsymbol{\rho}_{\mathrm{hh}}-\boldsymbol{\rho}_{\mathrm{ee}}\boldsymbol{\Delta}\right)\\ \circ & \circ\end{bmatrix}
\end{aligned}$$

D. Ultra-localized orbitals approximation

Let us now consider the explicit expression of the interaction term written in an arbitrary electronic basis $\{\phi_\alpha(\mathbf{r})\}$ in real space as

$$V_{\gamma\delta}^{\alpha\beta}=\int d\mathbf{r}d\mathbf{r}'\phi_\alpha^*(\mathbf{r})\phi_\beta^*(\mathbf{r}')V(\mathbf{r}-\mathbf{r}')\phi_\gamma(\mathbf{r}')\phi_\delta(\mathbf{r}) \tag{34}$$

where we assume $V(\mathbf{r})$ to be a generic electrostatic potential with symmetry $V(\mathbf{r}-\mathbf{r}')=V(\mathbf{r}'-\mathbf{r})$. In fact, it is from this very same symmetry that the relations we initially imposed in Eqs.(2) come from. Furthermore, consider that we are working with a crystalline system, such that we can specify one of the generic degrees of freedom to be a position vector in the lattice $\alpha=(\mathbf{R}_\alpha,\mathbf{s}_\alpha,\tilde{\alpha})$, with $\mathbf{R}_\alpha$ specifying the position within the Bravais lattice, $\mathbf{s}_\alpha$ the respective sub-lattice and $\tilde{\alpha}$ so we don't forget other generic degrees of freedom that the system might have. Moreover, if we consider a tight-binding (TB) framework and consider the orbitals to be ultra-localized, we can reasonably assume that $V(\mathbf{r}-\mathbf{r}')$ will not vary strongly within the orbital spread and thus, we could factor out $V(\mathbf{r}-\mathbf{r}')$ from the integral in Eq.(34), allowing us to write the interaction simply as

$$V_{\gamma\delta}^{\alpha\beta}\approx V(\mathbf{r}-\mathbf{r}')\int d\mathbf{r}\phi_\alpha^*(\mathbf{r})\phi_\delta(\mathbf{r})\int d\mathbf{r}'\phi_\beta^*(\mathbf{r}')\phi_\gamma(\mathbf{r}')=V(\mathbf{r}_\alpha-\mathbf{r}_\beta)\delta_{\alpha\delta}\delta_{\beta\gamma} \tag{35}$$

where we used the orthogonal property of the orbitals and with $\mathbf{r} \equiv \mathbf{r}_\alpha$ and $\mathbf{r}' \equiv \mathbf{r}_\beta$ being the centers of the orbitals α and β respectively. Within the scope of this approximation, the self-energies defined in Eqs.(23)-(25) now simply read as

$$\Sigma^H_{ab} \approx V(\mathbf{r}_a - \mathbf{r}_\gamma)\rho^{\mathrm{ee}}_{\gamma\gamma}\delta_{ab} \tag{36}$$

$$\Sigma^F_{ab} \approx -V(\mathbf{r}_a - \mathbf{r}_b)\rho^{\mathrm{ee}}_{ab} \tag{37}$$

$$\Delta_{ab} \approx V(\mathbf{r}_a - \mathbf{r}_b)\rho^{\mathrm{eh}}_{ab} \tag{38}$$

It is important to note that the Hartree term, contrarily to the Fock term, does not depend on the distance between the ultra-localized orbitals but rather on the purely electronic background density ρ_{ee}. One can interpret this as if the two orbitals would interact indirectly through the effects of the variations of the background density. In these sense, the Hartree term would already account for the screening dynamics of the system. This, however, cannot be said the same for the Fock term. Thus, if ones wishes to introduce some screening effect, one must add it manually; for example, by choosing, instead of a bare potential $V(\mathbf{r})$, an effective screening potential $V \to W(\mathbf{r}, \omega)$. Technically, to account for the dynamics of the screening, one would have to solve the BSE self-consistently since now the Fock self-energy itself would depend on the resulting excitonic dynamics. However, as discussed in the article Ref.[Ro00], taking the static limit W($\omega = 0$) is a reasonable approximation for most semiconductor crystals since the plasmon energies that actually control the dynamic of the screening are much bigger than the excitonic binding energies. Can I say this? Does this even make sense? I'm trying to justify the use of W in my Master's Thesis. Rui, on the other hand, wrote in his notes ¨The Hartree term is only sensible to the changes in the background density and thus, due to the large range of the screened Coulomb potential, can be at first order ignored.¨ I understand the concept of taking a mean over a large sample to ignore local variations but I don't understand why I can ignore the mean all together. What's even the meaning of "first order" in the statement. Regarding the paring term, one could also specify a different potential manually, for example, thinking of it as an on-site constant Hubbard-like term $V(\mathbf{r}) \to U < 0$.